

Project Proposal – Zurich Footfall Forecast

1 Problem Statement

Pedestrian traffic forecasting can be useful for retailers and city-center operators because pedestrian volumes vary strongly by hour, weekday, weather and seasonal patterns. The project focuses specifically on short-term pedestrian forecasting at “Bahnhofstrasse Mitte” in Zurich, using continuously updated real-world sensor data. The project uses historical observations from September 2021 onwards and focuses exclusively on the Bahnhofstrasse Mitte sensor location for the core project.

The goal is to predict hourly pedestrian measured during the corresponding hour on the following day. Model performance will be evaluated using Mean Absolut Error (MAE).

As a baseline the project will use a seasonal-naive forecast, which predicts that the pedestrian count for given hour will be equal to count observed at the same hour one week earlier. The baseline captures the strong weekly seasonality of pedestrian traffic while remaining simple and transparent.

The project will be considered successful if the final model archives MAE of 200 pedestrian or less on an out-of-time test set and improves the MAE of the seasonal-naive baseline by at least 20%.

2 Originality & Motivation

Pedestrian forecasting is relevant for retailers and city-center operators because pedestrian volumes can support short-term planning decisions. I chose this topic because I'm interested in time-series forecasting with continuously changing real-world sensor data. I reviewed previous HSLU MLOps projects and the KTH UD2223 project lists. While forecasting projects are common, I did not find a project specifically focused on pedestrian traffic prediction.

This project is therefore differentiated by using continuously updated pedestrian sensor data from Zurich to predict footfall at Bahnhofstrasse Mitte 24 hours ahead.

3 Data sources & features

The project uses the “Hystreet pedestrian frequency” dataset provided by the City of Zurich Open Data platform. The data is accessed through a public CSV download endpoint and therefore does not require web scraping. The source is updated hourly, while the feature pipeline will poll the dataset every three hours and ingest new or revised observations automatically.

The currently available dataset contains data from September 2021 onwards across four locations. For the core project only Bahnhofstrasse Mitte is used, providing approximately 43'800 hours observation at the moment. Since one new observation is produced per hour the selected location grows by 24 observations per day.

The prediction target is the pedestrian count 24 hours ahead at Bahnhofstrasse Mitte. The label therefore becomes available naturally when the corresponding future sensor measurement is published. Only verified observations will be used as training labels. Planned features include:

- Current pedestrian count
- Pedestrian count 1, 2 and 24 hours ago
- pedestrian count 7 days ago
- rolling pedestrian-count averages over previous time windows

- hour of day
- day of week
- weekend indicator
- month / seasonal information
- current temperature
- current weather condition

This is a regression problem, so the rare-positive class criterion is not applicable

To prevent data leakage, only information that would have been available at the prediction time will be used as a model input. In particular, future pedestrian counts or future observed weather values will not be used as features. Model evaluation will use an out-of-time split, where the model is trained on earlier periods and evaluated on later unseen periods instead of using a random train-test-split.

4 System design

The project follows a simple FTI (Feature-Training-Inference) architecture. The feature pipeline fetches live pedestrian data from Open Data Zurich, computes features and stores them. The training pipeline trains and registers the best model. The inference pipeline loads the latest model and features to generate a 24-hour-ahead forecast shown in simple web UI.

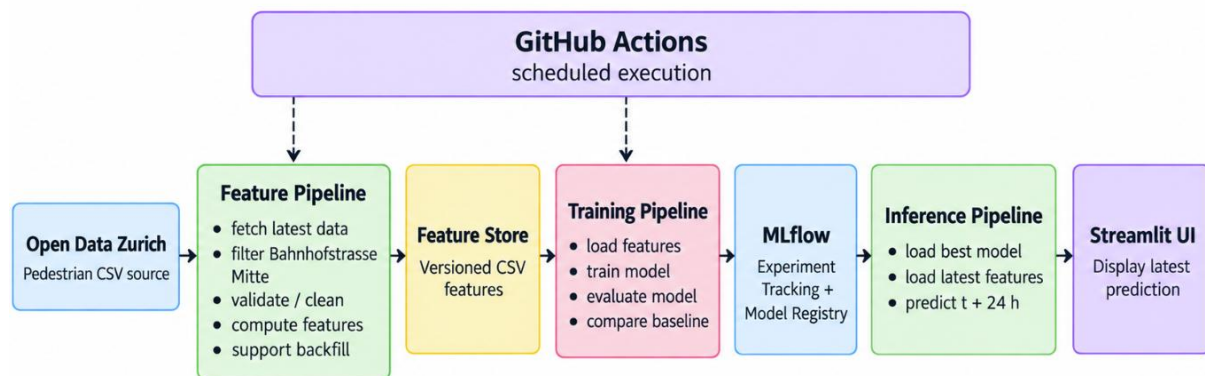


Figure 1: FTI architecture diagram

Component	Planned technology	Why
Feature Store	Versioned CSV	Simple and reproducible storage for engineered features
Experiment Tracking + Model Registry	MLflow	Tracks, runs and stores the best model
Orchestration	Github Actions	Runs ingestion and retraining on a schedule
Serving / UI	Streamlit	Display live predictions in a simple web interface
Reproducibility	Docker	Provides a consistent execution environment

Table 1: Planned MLOps Technology Stack

Optional stretch goals: Forecasting additional Bahnhofstrasse locations and adding advanced drift monitoring. These will only be implemented after the core FTI system is working reliably.

The GitHub Repository is publicly available at: <https://github.com/Deeliodas/MLOPS.git>